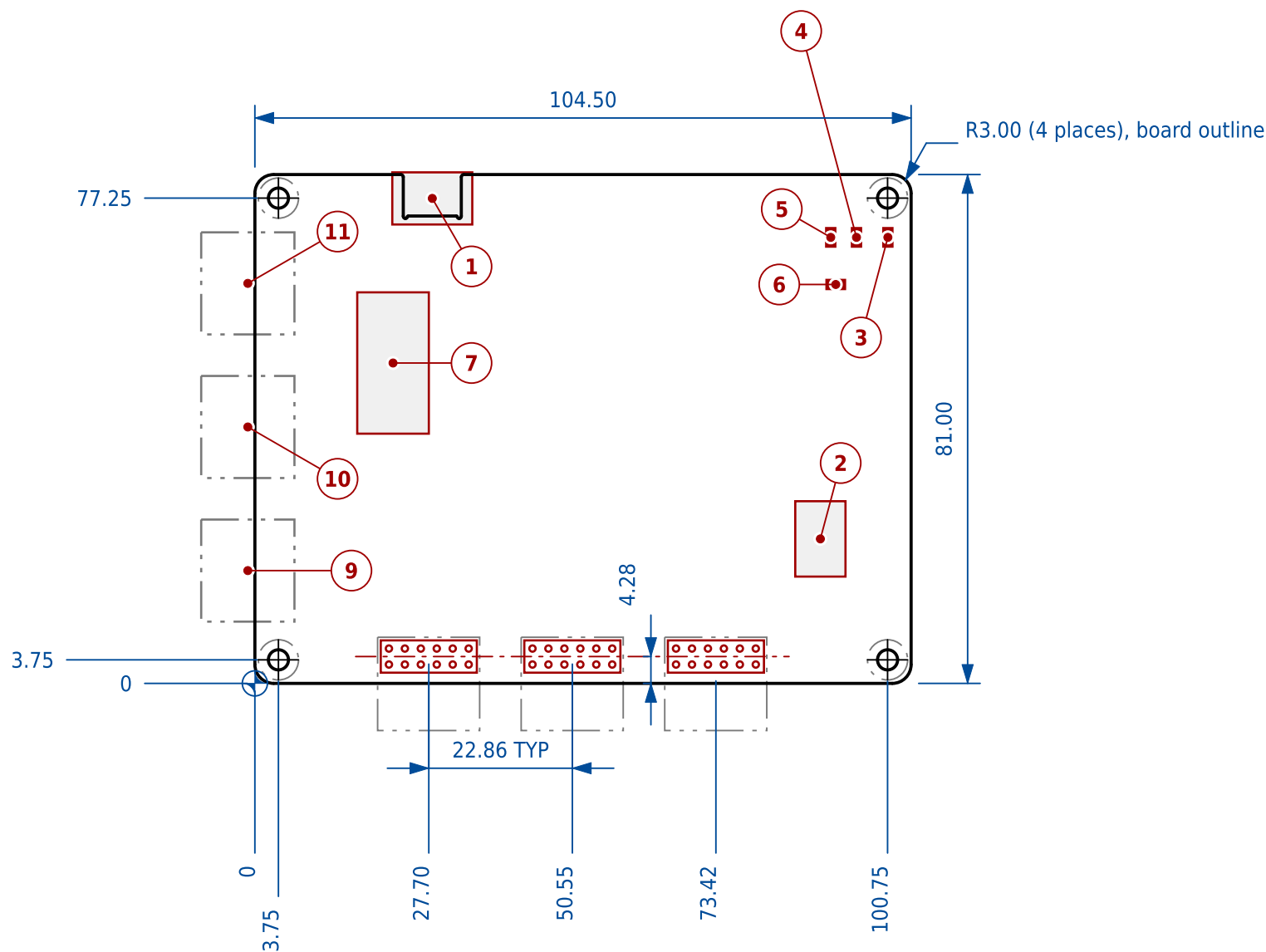




NOTES

1. Viewed from the component side.
2. Ships in TT03p5 ASIC IC kit, TT04 Dev Kit and TT05 Dev Kit.
3. Hole IDs are the KiCad reference designators.
4. TT03p5's board is recorded as rev 1.2.1 but unconfirmed: the TT03p5 render is labelled v1.1.2.
5. v1.2.2c is Tiny Tapeout's name for rev 1.2.3, an assembly variant of 1.2.2.
6. The revisions covered are mechanically identical; only the electrical design differs.
7. Pmod host pitch, 22.86 mm (0.9 in), is the Digilent Pmod Interface Specification 1.2.0 figure and is the same on every revision; the mounting plate, TT-MP-01, registers against it.
8. Assembled envelope, connector overhang included: 104.50 x 81.35 mm.

NOTES (continued)

9. The notch in the upper edge is a recess for the USB-C shell.

SOURCES

1. KiCad board file: https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb @ 8aad3f8 - title block: TinyTapeout 4+ Demoboard rev 1.2.1, dated 2024-04-12
2. KiCad board file: https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb @ cfdd80d7b - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26
3. KiCad board file: https://github.com/TinyTapeout/tt-demo-pcb/tinytapeout-demo.kicad_pcb @ a88cbc08b - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11
4. Shuttle mapping: https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrfFzF0urJLVWmdqAnrOJLetM/edit?gid=1250963370 - used by: TT03p5, TT04, TT05

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.75	3.75	3.20	6.40
MT2	100.75	3.75	3.20	6.40
MT3	3.75	77.25	3.20	6.40
MT4	100.75	77.25	3.20	6.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
INPUT	J3	bottom	27.695	4.275	34.045	5.545
BIDIR	J5	bottom	50.555	4.275	56.905	5.545
OUTPUT	J6	bottom	73.415	4.275	79.765	5.545

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C power / control	21.88 to 34.62	73.05 to 81.35
2	7-segment display	86.05 to 94.05	17.00 to 29.00
3	LED D1	100.07 to 101.53	69.52 to 72.48
4	LED D2	95.07 to 96.53	69.52 to 72.48
5	LED D3	90.97 to 92.43	69.52 to 72.48
6	LED D4	91.02 to 93.98	62.77 to 64.23
7	8-way input DIP switch	16.30 to 27.70	39.75 to 62.25
8	Second DIP switch - not on this board	-	-
9	Side Pmod J12 (not fitted)	-8.55 to 6.30	9.83 to 26.08
10	Side Pmod J13 (not fitted)	-8.55 to 6.30	32.69 to 48.94
11	Side Pmod J14 (not fitted)	-8.55 to 6.30	55.55 to 71.80

LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Pmod connector body, hole keep-out
- Dimension, extension and leader

DB 4+ V1.2.1 / DB 4+ V1.2.2 / DB 4+ V1.2.2C

tinytapeout-demo rev 1.2.1, 1.2.2 and 1.2.3

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-105-gb234968	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	TT-DB-02	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	